

Comparing Linear and Tree-Based Models for Food Delivery Time Prediction An Extension of My Previous Multilevel Modeling Project

1. Abstract

This project extends my previous analysis of food delivery time by comparing a simple linear regression model with two tree-based machine learning models: Random Forest and XGBoost. The goal was to test whether more flexible models could improve prediction by capturing nonlinear patterns and interactions among distance, preparation time, weather, traffic, time of day, vehicle type, and courier experience. Using an 80/20 train-test split, I evaluated all models with RMSE, MAE, and test-set R^2 . Contrary to my original expectation, the linear regression model performed best, with the lowest RMSE and MAE and the highest R^2 . The tree-based models were still useful because their variable importance results confirmed that distance and preparation time were the most important predictors. Overall, the results suggest that this simulated dataset is mostly explained by simple linear and additive relationships, so a more complex machine learning model was not necessary in this case.

2. Introduction

Accurately predicting food delivery time is an important problem for platforms such as DoorDash and Uber Eats because delivery-time estimates affect customer satisfaction, courier scheduling, and platform efficiency. In my previous project, I analyzed a publicly available Kaggle simulation dataset containing 1,000 delivery records. That analysis used linear regression and multilevel modeling to study how delivery time was associated with distance, preparation time, weather, traffic, time of day, vehicle type, and courier experience. The previous results showed that distance and preparation time were the strongest predictors, and that weather introduced modest group-level differences.

However, the previous modeling approach had two limitations. First, it treated distance as having a mostly linear relationship with delivery time. Second, it did not explicitly include interaction terms, such as whether the effect of distance changes under different weather conditions. Tree-based models can help explore these possibilities because they can capture nonlinear patterns and interactions automatically, without requiring each relationship to be manually specified.

This project therefore asks whether tree-based models improve prediction relative to a linear regression baseline, and whether they provide additional insight into the main factors associated with delivery time.

3. EDA and Methods

3.1 Data Preparation

The dataset contains 1,000 simulated delivery records with nine variables: distance, weather condition, traffic level, time of day, vehicle type, preparation time, courier experience, and delivery time. A missing-value check showed that the only variable with missing data was courier experience, with 30 missing values. These incomplete rows were removed, leaving 970 complete observations for modeling. Six extreme delivery times, ranging from 122 to 153 minutes, were identified by the IQR rule but retained because they corresponded to long-distance orders and may represent plausible delivery delays rather than data errors. The cleaned data were split 80/20 into training and test sets, with 778 observations in the training set and 192 observations in the test set.

3.2 Exploratory Data Analysis

Exploratory analysis showed three main patterns. First, delivery time is right-skewed, with most deliveries completed between about 40 and 80 minutes and a small number of much longer deliveries. Second, delivery time varies by weather condition, with snowy weather showing higher and more variable delivery times. Third, distance has a strong positive relationship with delivery time. The LOESS smoother suggests a mostly increasing pattern with some possible curvature at longer distances. These patterns motivated comparing a simple linear regression model with more flexible tree-based models.

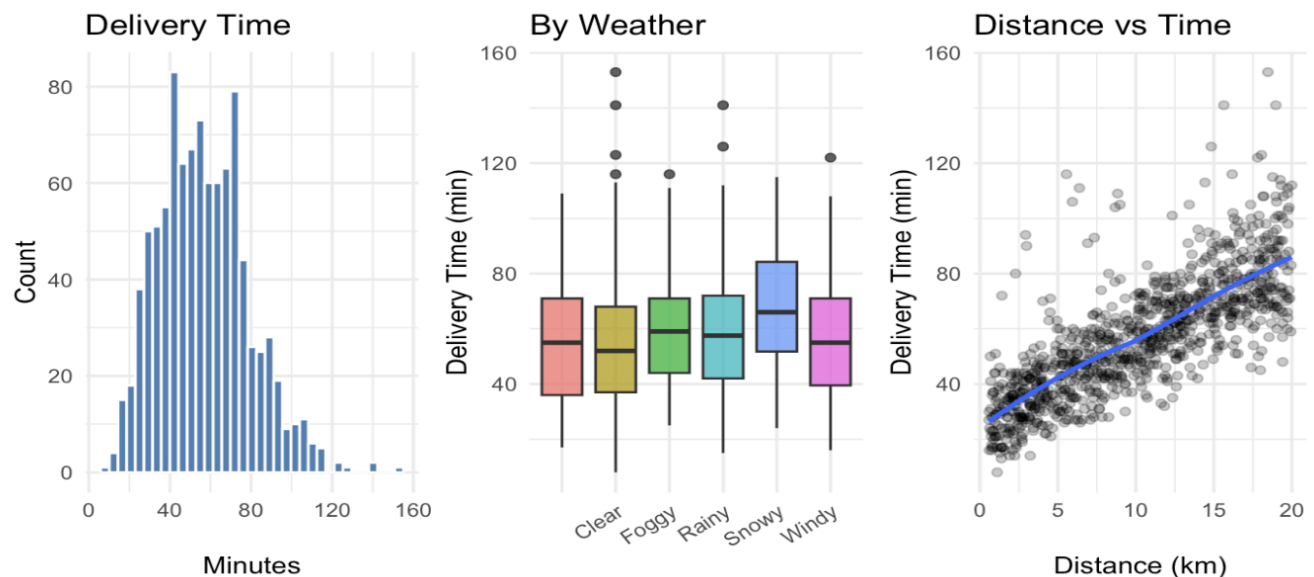


Figure 1: Exploratory analysis of delivery time. Delivery time is right skewed, varies across weather conditions, and increases strongly with distance.

3.3 Models

Baseline: Linear Regression. Linear regression was used as the baseline model because it is simple, interpretable, and closely related to my previous analysis. The model included distance, preparation time, courier experience, weather, traffic level, time of day, and vehicle type as predictors of delivery time.

Random Forest. Random Forest was used as a more flexible tree based model. It combines many decision trees and averages their predictions. This can help capture nonlinear relationships and interactions without manually specifying them. I used 500 trees and examined variable importance to understand which predictors contributed most to prediction.

XGBoost. XGBoost was used as a second tree based method for comparison. Unlike Random Forest, which averages many trees, XGBoost builds trees sequentially. Each new tree tries to improve the errors from the previous trees. Categorical variables were converted into dummy variables before fitting the model. A validation split within the training data was used for early stopping, and the held out test set was used only for final evaluation.

All models were evaluated on the same test set using RMSE, MAE, and test set R^2 . RMSE gives more weight to large prediction errors, MAE measures the average absolute prediction error in minutes, and R^2 measures the proportion of variation in delivery time explained by the model. Lower RMSE and MAE indicate better prediction, while higher R^2 indicates better overall fit.

4. Results

4.1 Predictive Performance

Table 1 summarizes the test set performance across the three models. Contrary to my original expectation, the linear regression model performed best, with the lowest RMSE, lowest MAE, and highest R^2 . Random Forest and XGBoost both had higher prediction errors and lower R^2 values than the linear model. This suggests that the main structure in this dataset is mostly linear and additive. Distance and preparation time already explain much of the variation in delivery time, so the extra flexibility of the tree based models did not improve prediction accuracy in this case.

Model	RMSE	MAE	R^2
Linear Model	10.424	6.459	0.760
Random Forest	11.055	7.564	0.730
XGBoost	11.348	7.503	0.715

Table 1: Test Set Model Performance

4.2 Variable Importance

Although the tree based models did not improve test set accuracy, their variable importance results were still useful for interpretation. Figure 2 shows the XGBoost variable importance by gain. Distance was by far the most important predictor, followed by preparation time. Weather, traffic level, and courier experience contributed much less, while time of day and vehicle type had very small contributions. This pattern supports the earlier regression results and suggests that delivery time in this dataset is mainly driven by trip distance and restaurant preparation time.

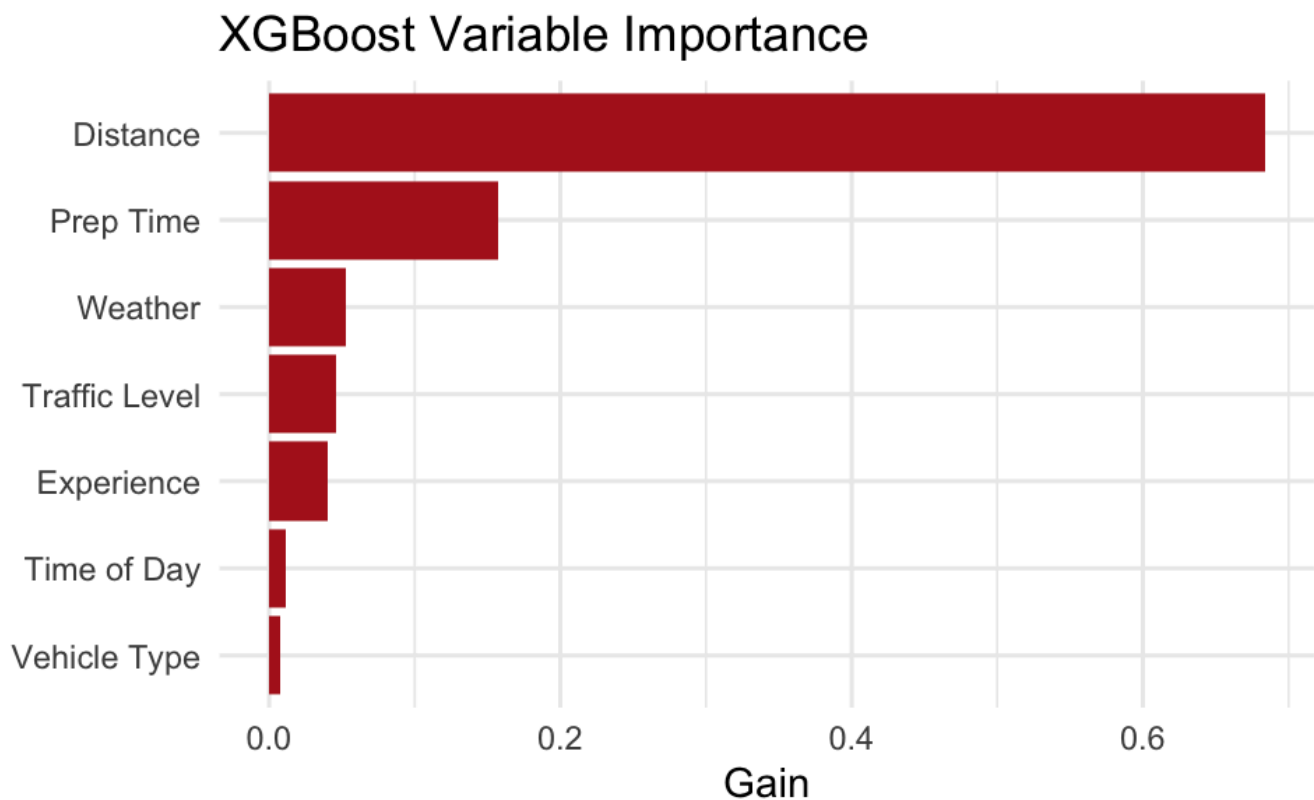


Figure 2: XGBoost variable importance by gain. Distance and preparation time were the dominant predictors, while other variables contributed much less.

4.3 Nonlinear Checks and Model Diagnostics

Partial dependence plots were used as exploratory checks to understand what the Random Forest model learned. The distance plot shows a strong increasing relationship between distance and predicted delivery time. The pattern is mostly linear, with some possible curvature at longer distances. The preparation time plot is also close to linear, which helps explain why the linear regression model performed well.

The distance and weather partial dependence plot suggests that snowy conditions are associated with higher predicted delivery times across distance values. However, this pattern should be interpreted cautiously because the tree based models did not improve test set accuracy compared with the linear model. Therefore, the partial dependence plots are useful for interpretation, but they do not provide evidence that the tree based models are better.

The predicted versus actual plots in the Appendix show that all three models perform reasonably well for moderate delivery times, but all models struggle with the longest deliveries. This is consistent with the right skewed distribution of delivery time observed in the exploratory analysis.

5. Discussion and Limitations

This project shows that more flexible machine learning models do not always outperform simpler statistical models. Although Random Forest and XGBoost can capture nonlinear patterns and interactions, the linear regression model achieved the best test set performance in this dataset. It had the lowest RMSE, lowest MAE, and highest R^2 . This suggests that the main signal in the data is mostly linear and additive.

The result is reasonable because distance and preparation time dominate the prediction problem. Once these two variables are included, the more flexible tree based models do not add much predictive value. The variable importance results still support the main conclusion from the previous analysis: delivery time is driven mostly by trip distance and restaurant preparation time, while weather, traffic level, courier experience, time of day, and vehicle type play smaller roles.

There are several limitations. First, the dataset is simulated, so it may be simpler than real food delivery data. Real delivery operations would include additional factors such as route type, restaurant workload, order size, courier availability, and real time traffic data. Second, the sample size is relatively small for flexible machine learning models. Third, partial dependence plots are useful for visualizing model patterns, but they are only marginal summaries and should not be interpreted as causal effects.

Future work could apply these methods to real platform data with richer predictors. It could also compare additional models such as GAMs or neural networks, or use a random slope multilevel model to study whether the effect of distance changes under different weather conditions.

6. Appendix

679 final project code

```
Rows: 1,000
Columns: 9
$ Order_ID          <int> 522, 738, 741, 661, 412, 679, 627, 514, 860, 13~
$ Distance_km       <dbl> 7.93, 16.42, 9.52, 7.44, 19.03, 19.40, 9.52, 17~
$ Weather           <chr> "Windy", "Clear", "Foggy", "Rainy", "Clear", "C~
$ Traffic_Level     <chr> "Low", "Medium", "Low", "Medium", "Low", "Low",~
$ Time_of_Day       <chr> "Afternoon", "Evening", "Night", "Afternoon", "~
$ Vehicle_Type      <chr> "Scooter", "Bike", "Scooter", "Scooter", "Bike"~
$ Preparation_Time_min <int> 12, 20, 28, 5, 16, 8, 12, 5, 20, 29, 13, 29, 12~
$ Courier_Experience_yrs <dbl> 1, 2, 1, 1, 5, 9, 1, 6, 6, 1, 4, 1, 5, 7, NA, 3~
$ Delivery_Time_min <int> 43, 84, 59, 37, 68, 57, 49, 46, 35, 73, 88, 76,~
```

Order_ID	Distance_km	Weather	Traffic_Level
Min. : 1.0	Min. : 0.590	Length:1000	Length:1000
1st Qu.: 250.8	1st Qu.: 5.105	Class :character	Class :character
Median : 500.5	Median :10.190	Mode :character	Mode :character
Mean : 500.5	Mean :10.060		
3rd Qu.: 750.2	3rd Qu.:15.018		
Max. :1000.0	Max. :19.990		

Time_of_Day	Vehicle_Type	Preparation_Time_min
Length:1000	Length:1000	Min. : 5.00
Class :character	Class :character	1st Qu.:11.00
Mode :character	Mode :character	Median :17.00
		Mean :16.98
		3rd Qu.:23.00
		Max. :29.00

Courier_Experience_yrs	Delivery_Time_min
Min. :0.000	Min. : 8.00
1st Qu.:2.000	1st Qu.: 41.00
Median :5.000	Median : 55.50

Mean	:4.579	Mean	: 56.73
3rd Qu.:	7.000	3rd Qu.:	71.00
Max.	:9.000	Max.	:153.00
NA's	:30		

Table 1: Missing value counts in raw dataset.

Variable	Missing
Courier_Experience_yrs	30

[1] 30

[1] 970

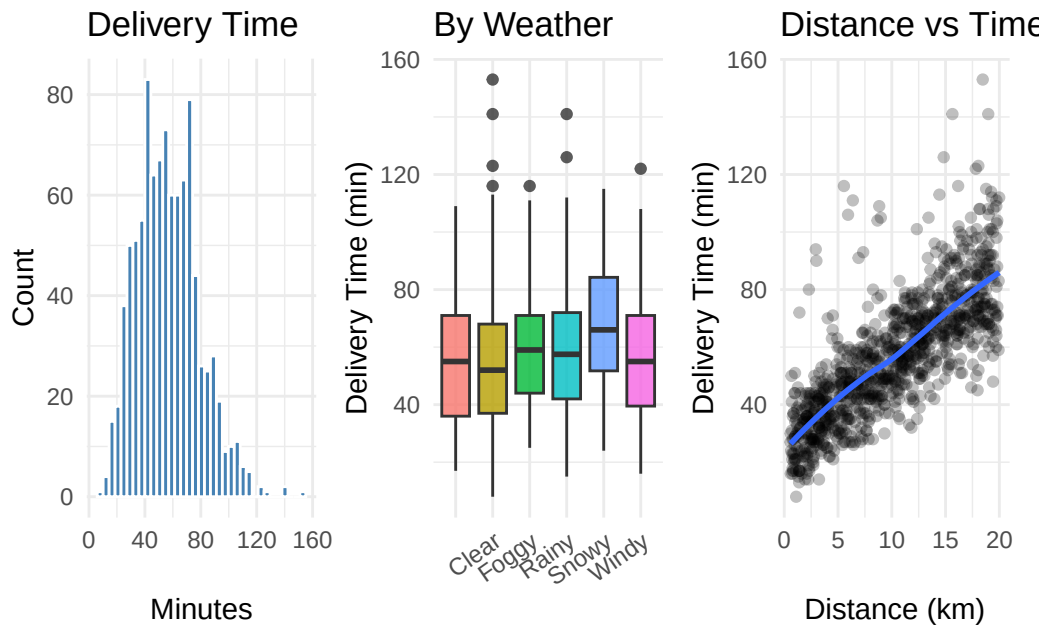
[1] 970

	Order_ID	Distance_km	Weather	Traffic_Level	Time_of_Day	Vehicle_Type
1	814	18.46	Clear			Scooter
2	446	18.97	Clear	Low	Evening	Car
3	394	15.64	Rainy	Low		Bike
4	385	14.83	Rainy	Low	Morning	Car
5	948	18.05	Clear	Medium	Evening	Scooter
6	428	17.81	Windy	High	Evening	Bike

	Preparation_Time_min	Courier_Experience_yrs	Delivery_Time_min
1	29		153
2	25		141
3	20		141
4	19		126
5	10		123
6	21		122

[1] 778

[1] 192



Call:

```
lm(formula = Delivery_Time_min ~ Distance_km + Preparation_Time_min +
    Courier_Experience_yrs + Weather + Traffic_Level + Time_of_Day +
    Vehicle_Type, data = train_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.932	-5.096	-1.140	2.810	61.794

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	26.90861	3.90805	6.885	1.21e-11 ***
Distance_km	2.97303	0.06668	44.585	< 2e-16 ***
Preparation_Time_min	1.00388	0.05386	18.638	< 2e-16 ***
Courier_Experience_yrs	-0.62652	0.13031	-4.808	1.84e-06 ***
WeatherClear	-6.17174	2.24814	-2.745	0.006189 **
WeatherFoggy	2.06233	2.48883	0.829	0.407571
WeatherRainy	-0.56217	2.33605	-0.241	0.809891
WeatherSnowy	3.49617	2.51406	1.391	0.164740
WeatherWindy	-4.24310	2.50805	-1.692	0.091096 .
Traffic_LevelHigh	2.86108	2.30307	1.242	0.214514
Traffic_LevelLow	-8.21047	2.21469	-3.707	0.000225 ***
Traffic_LevelMedium	-2.58020	2.21536	-1.165	0.244512


```

Time_of_DayAfternoon    -7.97355    2.20344   -3.619  0.000316 ***
Time_of_DayEvening      -7.39179    2.19991   -3.360  0.000818 ***
Time_of_DayMorning      -8.28838    2.20921   -3.752  0.000189 ***
Time_of_DayNight        -9.11375    2.47381   -3.684  0.000246 ***
Vehicle_TypeCar          -0.14927    1.02010   -0.146  0.883702
Vehicle_TypeScooter     -0.43201    0.88000   -0.491  0.623625

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.56 on 760 degrees of freedom

Multiple R-squared: 0.7801, Adjusted R-squared: 0.7752

F-statistic: 158.6 on 17 and 760 DF, p-value: < 2.2e-16

Call:

```
randomForest(formula = Delivery_Time_min ~ Distance_km + Preparation_Time_min + Courier_Type, data = data, ntree = 500, importance = TRUE)
```

Courier_Type

Type of random forest: regression

Number of trees: 500

No. of variables tried at each split: 3

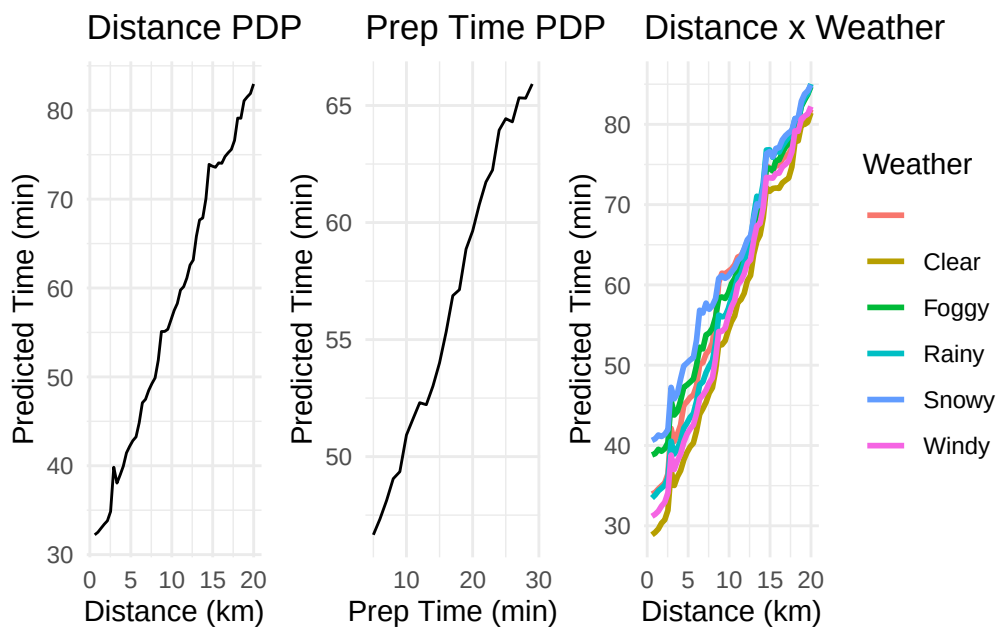
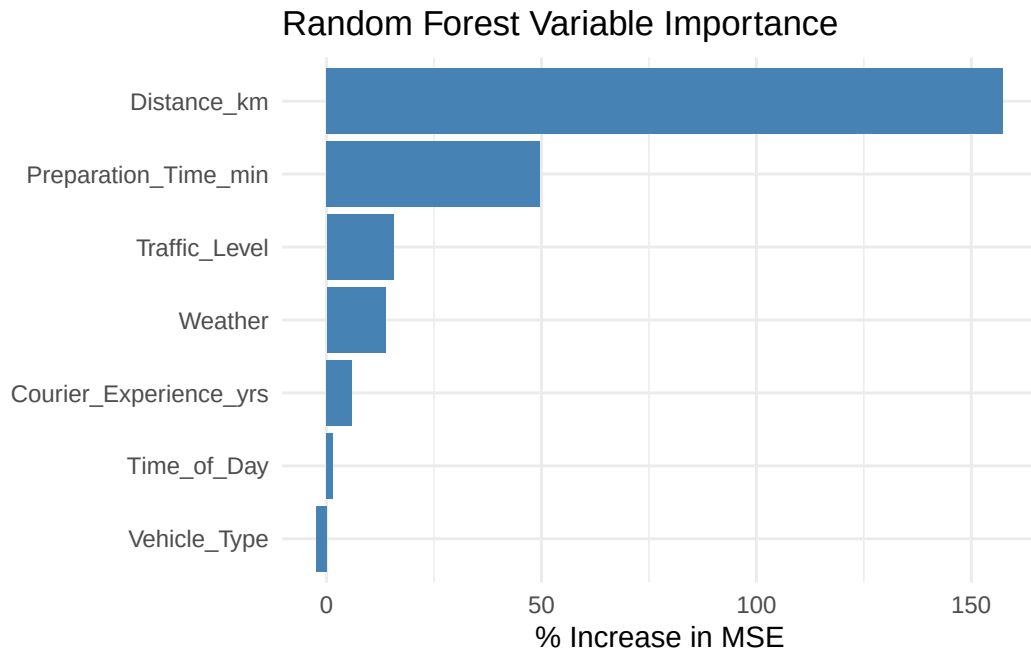
Mean of squared residuals: 135.7945

% Var explained: 72.58

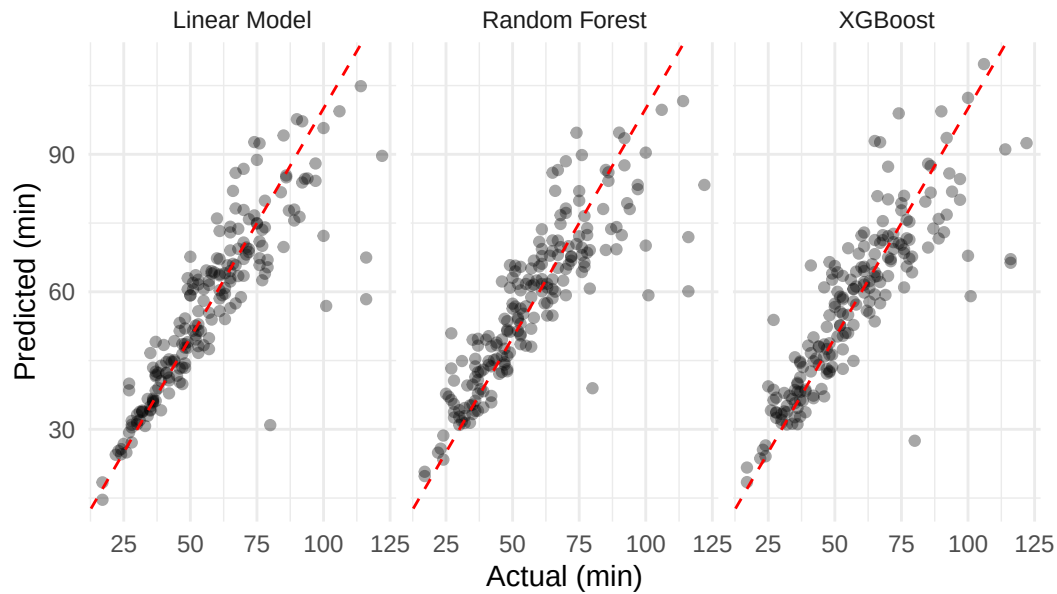
NULL

Table 2: Out-of-Sample Model Performance

Model	RMSE	MAE	R2
Linear Model	10.424	6.459	0.760
Random Forest	11.055	7.564	0.730
XGBoost	11.348	7.503	0.715

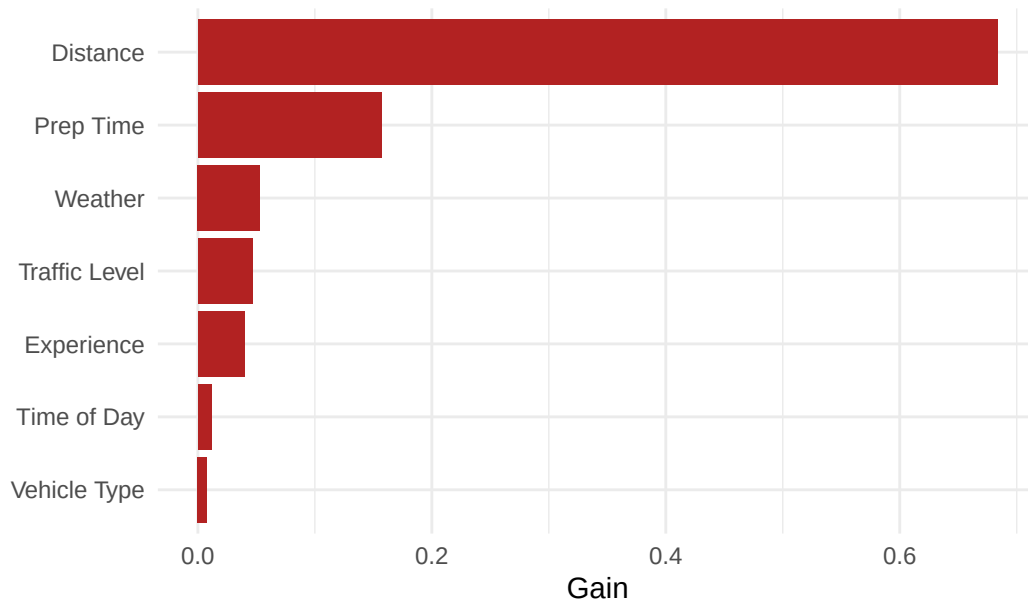


Predicted vs Actual Delivery Time



	Feature <char>	Gain <num>	Cover <num>	Frequency <num>
1:	Distance_km	0.683676860	0.412606683	0.31080438
2:	Preparation_Time_min	0.157262063	0.209630077	0.22941456
3:	Courier_Experience_yrs	0.039928705	0.106241992	0.13993337
4:	Traffic_LevelHigh	0.025660442	0.039886928	0.03950500
5:	WeatherClear	0.018943893	0.035714528	0.03807711
6:	Traffic_LevelLow	0.018776110	0.043574639	0.03950500
7:	WeatherFoggy	0.012255689	0.020567121	0.02950976
8:	WeatherRainy	0.011223114	0.022014412	0.02332223
9:	WeatherSnowy	0.009046021	0.022221168	0.02284626
10:	Vehicle_TypeCar	0.004428473	0.012188426	0.02094241
11:	Time_of_DayEvening	0.003659336	0.007280517	0.01808663
12:	Time_of_DayAfternoon	0.003636621	0.011446138	0.01761066
13:	Vehicle_TypeScooter	0.003418028	0.008338022	0.01713470
14:	Time_of_DayMorning	0.002964137	0.009500600	0.01237506
15:	Traffic_LevelMedium	0.001978652	0.006938183	0.01094717
16:	WeatherWindy	0.001685509	0.018279249	0.01475488
17:	Time_of_DayNight	0.001456349	0.013571317	0.01523084

XGBoost Variable Importance



Random Forest OOB RMSE

